

DRYLANDS SUMMER SCHOOL

2nd Drylands Summer School / Concept Note

Monitoring Restoration in Agrosilvopastoral Systems for Sustainability and Ecosystem Services

Nairobi, Kenya – 12-15 September 2024



Context

Drylands and rangelands sustain over 2 billion people globally, but face challenges from conversion for agricultural and other land use, climate change, inappropriate restoration and afforestation programmes, and unsustainable practices. The dryland programme of the Food and Agriculture Organization of the United Nations (FAO) aims to understand and promote good practices for the protection, sustainable management, and restoration of drylands.

Similarly, the United Nations International Year of Rangelands and Pastoralists (IYRP) 2026, initiated by Mongolia and supported by the European Union and 60 Member States, with FAO as lead agency, aims to raise awareness and increase responsible investment in the pastoral livestock sector, including sustainable land management, ecosystem restoration, and equitable access to markets. In its 26th session, the FAO Committee on Forestry (COFO) recognized the progress of the COFO Working Group in Dryland Forests and Agrosilvopastoral Systems (COFO WG) and urged it to support the IYRP through cross-regional expertise exchange.

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Drylands encompass wide-ranging environmental and socio-economic differences. Governance structure, political stability and therefore sustainability choices may greatly vary across regions and countries, and a context-dependent approach to building the resilience of dryland production systems is needed. As part of this, the COFO WG is working to build resilient drylands by training experts to advocate for transformative dryland management. Those trained experts can facilitate the creation of a shared vision and create a wave of transformation in dryland policymaking and management. The inaugural Drylands Summer School in 2023 in Amman, Jordan, partnered with the Agricultural Research Council of South Africa, Jordan University of Science and Technology, the Sokoine University of Agriculture in Tanzania, Swedish University of Agricultural Science, the University of Jordan, and the World Overview of Conservation Approaches and Technologies (WOCAT). The four-day course empowered the 22 participants from all dryland regions with tools to enhance dryland resilience, emphasizing context-specific solutions and defining cross-cutting indicators for monitoring progress.

2nd Drylands Summer School

The Second Drylands Summer School will be held in partnership with the United Nations Convention to Combat Desertification's (UNCCD) G20 Global Land Initiative, the Community Jameel, the Center for International Forestry Research and World Agroforestry (CIFOR-ICRAF), the World Overview of Conservation Approaches and Technologies (WOCAT) whose Secretariat is hosted by the Centre for Development and Environment (CDE) of the University of Bern, The Swedish University of Agricultural Sciences (SLU), the Jameel Observatory, the Agricultural Research Council of South Africa (ARC-LNR), the International Year of Rangelands and Pastoralists (IYRP) 2026, the GEF-7 Sustainable Forest Management Impact Program on Dryland Sustainable Landscapes (DSL-IP), and the Kenya Forest Service. As well as being a partner, the UNCCD G20 Global Land Initiative is the main sponsor of the Summer School and the Community Jameel funds the field trip.

The Summer School will take place at the CIFOR-ICRAF Campus in Nairobi, Kenya from Thursday 12 September to Sunday 15 of September. The programme includes a field trip on Saturday 14 September.

Objectives

The overall objective of the 2nd Drylands Summer School is to strengthen capacities to follow-up, monitor, and provide feedback on the actions and related issues in sustainably managing dryland forests and agrosilvopastoral resources, with a special focus on pastoralism-based systems. It will further build capacities of experts and practitioners on the transformational approach to dryland resilient systems, with a special focus on **Monitoring Restoration in Agrosilvopastoral Systems for Sustainability and Ecosystem Services**. The specific objectives are:

1. **Understand Challenges and Approaches:** a thorough understanding of the challenges in sustainable dryland management and participatory approaches for monitoring restoration interventions in agrosilvopastoral systems.

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2. **Review and Apply Monitoring Frameworks:** Adapt, modify and (re)-apply different frameworks for monitoring and providing feedback on drylands projects, focusing on sustainable management, adaptation, and restoration.
3. **Co-define Indicators and Gather Data:** Collaborate with local stakeholder to define cross-cutting sets of indicators and gather field data.
4. **Share Work and Build Community:** Present and exchange participants' own work on monitoring drylands with peers and initiate the building blocks for a long-term community of practice.

Target Participants

Target participants include experts, practitioners, project managers, M&E specialists and policy makers who have responsibilities in the management of agrosilvopastoral systems within dryland areas.

Partners will widely disseminate an open call for applications, and 20-25 participants will be selected based on motivation, capacities, and potential impact, whilst considering gender, geography, and other aspects of diversity.

Participants are required to complete the FAO [Transforming dryland forests and agrosilvopastoral systems course](#), (3h 49m duration), and prepare a generic case study, based on their own work, to capture experiences in the management of agrosilvopastoral systems. A template for their case study will be provided upon selection.

Programme Structure

The Drylands Summer School is a professionally facilitated course combining theoretical knowledge with practice-based learning, including group discussions, peer-to-peer learning, field visits and technical lectures. The course is built on the principles of experiential learning, which places the participants' knowledge and experience at the centre. Participants will share and critically reflect on their work to develop a deeper understanding of their own projects while learning how to improve the participatory monitoring of their restoration interventions.

The Summer School will last four days, including a field trip. The entire course will be delivered in English. Professional facilitation and learning support will be provided by a team of CIFOR-ICRAF dryland restoration experts and specialists from FAO and other partners. The Summer School will focus on experience-sharing and networking, to ensure that the group stays connected to a growing network of dryland restoration professionals across the world. Support to the network will be provided for continuous post-Summer School engagement, through a digital platform in close collaboration with willing partners.

A digital badge certification will be provided upon completion of the Summer School.

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[Detailed Programme](#)

Learning Objectives:

1. Understand Challenges and Solutions for Drylands and Why to Monitor them

- Perform a thorough understanding of the challenges in sustainable land management and restoration initiatives in dryland forests and agrosilvopastoral systems. Analyse the monitoring needs and the different roles involved. Establish a clear link between participation and monitoring.

2. Assess and Implement Monitoring Frameworks

- Review available options and choose, adapt, modify and apply a framework for monitoring, evaluating and providing feedback on drylands initiatives.

3. Gather field data and Co-define Indicators for Effective Monitoring

- Develop participatory processes involving local stakeholders to define cross-cutting sets of indicators and gather field data.

4. Share Practice, Build Community and Contribute to Enabling Environments

- Disseminate the monitoring work, provide pathways for improving monitoring in drylands, present and exchange experiences on monitoring drylands and initiate the building blocks for a long-term community of practice.

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Topics per day:

Day 1: Introduction and principles of monitoring drylands • Welcome and opening • Keynote presentation – setting the scene on dryland management • Introduction, principles and practices of monitoring drylands • Monitoring in practice • Getting to know each other • Presentation of individual case studies and action plan • Group work on selected case studies	Day 2: Monitoring frameworks and sustainability indicators • Participatory monitoring • Monitoring frameworks • Use of indicators to monitor drylands initiatives • Group work on selected cases • Individual work on action plan
Day 3: Monitoring in practice Field trip - TBD	Day 4: Scaling-up from site experience to planet surveillance • Citizen data collection in practice • Scaling-up, from local to global scales • Individual action plan & shared vision • Building blocks for a long-term community • Next steps, evaluation & closing

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Programme

DAY 1: Thursday 12th of September

Understanding challenges in sustainable dryland management and approaches to monitoring dryland degradation and restoration processes

Time	Activity	Description	Who
08.45 – 09.00	Registration		Team
9:00-9:45	Welcome and opening	Opening	Eliane Ubalijoro, CEO, CIFOR-ICRAF and Director-General, ICRAF
		Welcome	Tiina Vähänen, Deputy Director, Forestry Division, FAO
		Introduction to Summer School Programme and expected results	Thomas Hofer, Team leader, Forests and Biodiversity, Forestry Division, FAO
		Learning objectives Methodology: activities, group work, personal work, presentations, discussion, field trip Role of individual case studies and action plans	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team
		Logistics, security, essential tips	Peter Murunga, CIFOR-ICRAF Security Officer and FAO Kenya Security Officer
9:45-10:15	Keynote Presentation 1) Setting the scene (from policy to action) 2) Challenges and opportunities of drylands	Introduction to drylands as social-ecological systems (link and recap e-course) Overview of challenges and opportunities in drylands	Igshaan Samuels, ARC, Co-chair of 2026 IYRP



Food and Agriculture
Organization of the
United Nations



United Nations
Convention to Combat
Desertification



G20 GLOBAL
LAND INITIATIVE



JAMEEL
OBSERVATORY
FOR FOOD SECURITY EARLY ACTION



WORKING
GROUP
ON DRYLANDS



CIFOR



World
Agroforestry



SLU



ARC • LNR



WOCAT^u



IYRP
2026



Community
Jameel

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	3) Sustainably managing agrosilvopastoral systems	Discussion on sustainability and transformation in dryland agrosilvopastoral systems	
10:15-10:45	Keynote Presentation 1) The importance of monitoring 2) Global initiatives and monitoring activities	Global initiatives, platforms and tools on monitoring drylands	FAO team, with a short statement by UNCCD secretariat
10:45-11:15	Coffee Break		
11:15-11:30	Getting to know each other	Exercise - TBD	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team
11:30-12:30	Participants Case Presentations Selection of 1 case per group for group work	Creation of small work 4-people groups. 6 groups Each participant presents their own case study to a small group (10 minutes/participant) Select 1 case for each group. Initial discussion on management and restoration objectives and monitoring challenges	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team Partners facilitate and take notes on each group
12.30-13.30	Lunch Break		
13:30-14:00	Session 1: Monitoring Land Restoration in Drylands - Modern Approaches	The G20 perspective on monitoring drylands for ecosystem restoration, in theory and practice.	UNCCD G20 team
14:00-15:00	Landscape mapping exercise	Participatory drawing exercise to better understand the selected cases, their objectives and monitoring needs. Identification of challenges, stakeholders and ongoing projects/activities. Presenting landscape maps to other groups	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team) /Partners collaboration

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15:00-15:30	Coffee Break		
15:30-17:00	Session 2: Overview of tools and practices for monitoring dryland use systems	Monitoring and evaluation tools for dryland forests and agrosilvopastoral systems. Drylands monitoring: Principles and approaches: (Getting the Idea, options, tools, what to take into account). PMERL Participatory processes for monitoring drylands. The role of facilitators (with an invitation to participants to reflect in their own roles)	FAO Team
17:00-17:30	Group work: monitoring frameworks	Selected case studies: examples of successful participatory monitoring frameworks, what works/doesn't work, Exercise TBD	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
17:30-17:45	Personal work on the Individual action plan	Participants get time to write down their lessons learned of the day in their individual action plan	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
DAY 2: Friday 13th of September <i>Review and adapt frameworks for monitoring of drylands projects</i>			
09:00-09:30	Daily reflection exercise		Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team, /Partners collaboration
09:30-11:00	Session 3: Monitoring drylands restoration processes	Monitoring dryland restoration and degradation processes. Use of the SLM PC (and integrated FAO-WOCAT tools such as the QM) in the Karoo, Eastern Cape and the Olifants landscapes in South Africa.	Lehman Lindeque, WOCAT
11:00-11:30	Coffee break		

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11:30-12:30	Adapting and modifying monitoring frameworks (+case work in groups) have another wrap up of the exercises	Group work exercise: Review and adapt frameworks to Participants' Cases Groups present their adapted frameworks. Peer review and facilitator feedback	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
12.30-13.30	Lunch Break		
13:30-15:00	Session 4: Defining and using indicators for monitoring degradation and restoration processes in drylands	Definition and use of indicators, linked to monitoring goals and objectives How to develop indicators Existing sets of indicators, where to look (e.g. ICRAF's Land Degradation Surveillance Framework, LDSF[ii]). Quality and selection criteria	Aida Bargues, SLU, Leigh Winowiecki and Christine Magaju, CIFOR-ICRAF
15:00-16:00	Exercise case work in groups – TBD	Definition of indicators for monitoring group and individual case studies.	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
16:00-16:30	Coffee Break		
16:30-17:00	Session 5: Monitoring pastoralist activity	Approaching pastoralist communities under a monitoring perspective	Guyo Roba, CEO Jameel Observatory / ILRI, Kenya
17:00-17:45	Preparation for field trip & logistics	Programme and itinerary, Field trip assignment, Logistics	FAO / Partners
17:45-18:00	Individual action plan	Participants get time to write down their lessons learned of the day in their individual action plan	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
DAY 3: Saturday 14th of September			
<i>Learning from the field: Field Trip to Local Dryland Restoration Site</i>			

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	Field trip is to a site where participatory monitoring has been implemented so the processes/framework followed can be discussed at the site Participants reflecting on actual monitoring and planning what mechanisms they can incorporate what not to their own cases	Example of participatory monitoring of dryland restoration in practice How is the framework set up, how are indicators developed? How do local stakeholders experience this form of monitoring? Which techniques for data collection and analysis are used? What is the role of project manager/facilitator What works/does not work?	FAO / Partners
DAY 4: Sunday 15th of September <i>Developing action plans for participatory monitoring</i> Closing remarks and farewell			
09.00-09.30	Reflection exercise	Reflection on field trip	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
10:00:11:00	Session 6. Citizen data collection & participatory monitoring. The Regreening App	Practical training on the Regreening App on campus (mobile app for assisted citizen science data collection of dryland restoration activities)	Benard Onkware, Muhammad Ahmad, CIFOR-ICRAF, Aida Bargues, SLU
11:00-11:30	Coffee Break		
11:30 12:30	Session 7: Scaling-up monitoring	Combining citizen's data collection, participatory processes, systematic field surveys, and Earth Observation from local to global scales	Tor Vågen, Leigh Winowiecki, CIFOR-ICRAF, Aida Bargués, SLU, FAO and WOCAT All partners
12.30-13.30	Lunch Break		

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13:30-14:30	Individual action plan & shared vision	Participants get time to consolidate their lessons learned and develop concrete action points upon return Shared vision in pairs to provide each other feedback on their action plan	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
14:30-15:00	Building Blocks for a Long-term Community	Brainstorming session on how to stay connected and maintaining collaboration. Towards a community of practice on drylands	FAO team
15:00-15:30	Diving into the future	Ideas for the future, where to engage, challenges and opportunities: IYRP, next events, next summer school...	Igshaan Samuels, ARC, Co-chair of IYRP
15:30-16:00	Coffee Break		
16:00-17:30	Closing Remarks	Take home message – exercise Summary, conclusions, and evaluation	Cora van Oosten & Ilse Hennemann, CIFOR-ICRAF team /Partners collaboration
17:00-17:30	Reflection on the Summer School		Thomas Hofer, Team Leader, Forests and Biodiversity, Forestry Division, FAO
17:30-18:00		Closing and Farewell	Tiina Vähänen, Deputy Director, Forestry Division, FAO
DAY 5: Monday 16th of September Optional activity day (TBC)			

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Follow-up

The Drylands Summer School series is a unique opportunity to bring together participants to develop a facilitated network such as a community of practice.

The GEF-7 Sustainable Forest Management [Impact Program on Dryland sustainable Landscapes](#) (DSL-IP) has established two key communities of practice in close partnership with WOCAT and IUCN, and leverages on the Nurturing Community of Knowledge Practice for Women in Dryland Forests and Agrosilvopastoral Systems (WeCaN) and Making Every Voice Count for Adaptive Management (MEV-CAM) initiative for knowledge exchanges. Practitioners can engage and exchange lessons learned and experience, including Dryland forests and agrosilvopastoral systems monitoring by accessing [DSL-IP Communities of Practice](#).

Summer School participants are also welcome to join the DSL-IP e-Learning Series, which will address key aspects of the sustainable management of Dryland forests and agrosilvopastoral system. [Stay connected](#).

The COFO WG and FAO's Drylands team are currently planning a follow-up that began with the 1st Summer School, asking prior participants for feedback to share online.

The 2nd Drylands Summer School aims to amplify this network, through the following steps:

- 1) Report the results of the 2nd Summer School in GLF-Africa, programming a facilitated mini-workshop with Summer School participants.
- 2) Contact participants for feedback and information to share with wider network and to create mailing list.
- 3) Send 3-4 updates a year to all participants.
- 4) Include interested participants in the MEV-CAM community of practice and other networks linked to the Summer School partnership.

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Roadmap

2 nd Summer School / 2024	Apr	May	Jun	Jul	Aug	Sep	Oct
Concept note							
Partners' meeting							
Mini Workshop on program							
Call for applicants							
Resource mobilisation							
Applicant selection							
Agenda							
Program completed with speakers							
Completed logistics and materials							
Celebration 2 nd Drylands Summer School							
Reporting							

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Partnership:

Partners / Contribution	Budget	Teaching	Venue	Participants	Field trip	Design and contents
FAO - Drylands (Pedro / Victoria / Thomas / Marcelo)		2 people, full expenses covered		TBC		Lead Partner
UNCCD G20	USD 21000	1 person, full expenses covered		6 people sponsored, committed USD 18000		Partner, Key fund provider
FAO – Kenya			Additional support			
CIFOR-ICRAF (Cora, Ilse)		1 person, expenses pending on budget, facilitation team, other 2 people involved (Upon budget availability)	CIFOR-ICRAF Campus,			Partner
WOCAT (Nicole, Rima, Lehman)		1 person, full expenses covered				Partner
SLU (Aida)		1 person, full expenses covered				Partner
IYRP (Igshaan)		1 person, full expenses covered		2-3 participants from South Africa sponsored		Partner
IUCN (Chris/Bora)		1 person, full expenses covered				Partner
Jameel Observatory / Community Jameel	USD 1400	1 person, full expenses covered			Cover expenses bus and catering	Partner and financial support provider

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Tentative budget

This is the budget agreed at 03/05/2024 between FAO and CIFOR for the venue and basic logistics. Further resource mobilization is needed to secure all the Summer School needs. This is the main destination of the actual budget provided by the G20.

Item	3-days Summer School/Total (USD)	Notes
Meeting Room (conference hall)	1.500	Maximum for 3 days in Conference room
Tents & Chairs	3.000	(estimated cost of tent for the week where lunch is served)
Morning and afternoon tea/coffee with assorted pastries and mini sandwiches, Lunch	6.000	Morning and afternoon tea/coffee with assorted etc and Lunch
Facilitator	2.600	Four days professional facilitation
Travel costs resource persons	4.000	
Sub-total	17.100	
Management fee 21%	3.591	
Contingency 10%	1.710	
Grand total	22,401	

Additional costs for the Field Trip will be covered by Community Jameel, including hiring a bus with driver and providing lunchboxes for all participants.